

# SDN Branch Work Report

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Factory Flow (AECE Checkpoint)

**Author:** Shaun Bennet (stopdoingnothing)

**Period:** April 1-8, 2026

**Commits:** 67

**Scope:** Infrastructure, hardening, compliance, RBAC,  
leave engine, theme system, deployment tooling

## Feature Overview

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### Core Domain: Leave Management

- 1. BCEA Leave Accrual Engine** — Full South African labour law compliance. Monthly pro-rated annual leave accrual, graduated sick leave in the first 6 months with transition to full entitlement, family responsibility leave gating, accrual-pausing leave types, carry-over with configurable grace/forfeiture, backdated backfill, and termination settlement.
- 2. Custom Leave Rules Engine** — Configurable rules for non-BCEA types (study leave, religious leave, etc.) with phase-based entitlement that activates at tenure milestones. Supports 4 accrual modes: per-days-worked, monthly, annual, and fixed-per-cycle.
- 3. Future Leave Projection** — Advisory engine that simulates future balances month-by-month without DB writes. Detects cycle resets, calculates carry-over, and settles pending requests. Powers a progressive UI that shows current balance immediately and refines with projected values.
- 4. Half-Day Leave** — AM/PM toggles on leave requests, reducing totals by 0.5 days each. Carries through to settlement.
- 5. Server-Side Leave Hardening** — Statutory type enforcement, unpaid leave 7-day notice, medical certificate triggers/enforcement, overlap detection, 8-hour escalation reminders, optional HR approval stage, and an HR reporting endpoint.

### Access Control

- 6. Multi-Role RBAC** — Replaced legacy single-role fields with additive `roles[]` array (`employee`, `manager`, `hr`, `admin`). Migrated across all auth, leave, org chart, and profile code. Passwords stripped from API responses.
- 7. Configurable Role Permissions** — Runtime permission map stored in settings. `canSee()` / `canDo()` hook with admin bypass. Checkbox grid UI for configuration. Guards dashboard nav items and personnel actions.

### Infrastructure

- 8. Docker Deployment** — Multi-stage Dockerfile, 3 deployment profiles (dev with live reload, UAT with isolated ports/data, production with pre-built GHCR image). PostgreSQL 16 with health checks. Entrypoints handle migration, table pre-creation, and startup.
- 9. Deployment Tooling** — Makefile (setup, start, stop, update, backup, restore, reset), first-run setup.sh wizard, `.gitattributes` for LF enforcement, CRLF stripping in Docker build, raw SQL migrations via psql.
- 10. Backup & Bootstrap** — Automated 6-hourly `pg_dump` with 90-day retention. JSON export/import with bootstrap detection for fresh deployments. Selective exclusion of face descriptors, attendance photos, audit logs, and sessions. Idempotent import in dependency order with date string revival.

### Observability & Quality

- 11. Audit Log** — JSONB diff tracking for user profile changes, leave balance adjustments, approval decisions, and settings mutations. Actor, entity, and timestamp indexed.
- 12. In-App Feedback** — Bug report / feature request modal with localStorage queueing, exponential backoff retry, and Agent Gateway integration that creates GitHub Issues. Graceful degradation when gateway is unavailable.

### UI/UX

- 13. Semantic Theme Tokens** — All hardcoded palette colors migrated to HSL CSS custom properties with light/dark

variants. Industrial aesthetic (dark red, black, greys) with status color system and utility classes.

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## Context

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The Factory Flow (AECE Checkpoint) codebase was originally developed on Replit under the [quanga01](#) author (362 commits, Dec 4 2025 – Mar 31 2026). On April 1 2026 the repository was brought onto the stopdoingnothing profile and development continued under [Shaun Bennet](#). This report covers the 67 commits contributed during that period.

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## Day-by-Day Breakdown

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### Day 1 — Apr 1: Infrastructure & Leave Hardening (18 commits)

#### Docker & Deployment

- [e842a97](#) — Docker setup for on-prem deployment (Dockerfile, compose files, entrypoint)
- [4833290](#) — README with setup and deployment instructions
- [26a214d](#) — Simplified README (SESSION\_SECRET optional)
- [f361554](#) — Outstanding infrastructure and schema changes committed

#### Leave System — Server-side Hardening

- [09fc709](#) — Server-side leave validation and statutory leave types
- [cef85ba](#) — Sick leave probationary rule rewritten to strictly follow BCEA Section 22
- [d6257a3](#) — Unpaid leave 7-day notice validation with discretion bypass
- [824cf08](#) — Medical certificate flag triggers on sick leave submission
- [0e7bffc](#) — Custom leave rule execution engine
- [c732e32](#) — Enforce medical certificate upload before HR approval of sick leave
- [3b572ff](#) — Carry-over grace period made configurable via settings
- [cafe3e1](#) — HR approval stage made optional via setting
- [ba10fa7](#) — HR leave reporting endpoint ([GET /api/reports/leave](#))
- [441f93a](#) — Fixed inflated leave balance totals

#### Security & Compliance

- [cd1e1e4](#) — Server-side session auth enforcement
- [32ab5d9](#) — Audit log for sensitive admin mutations
- [0df1953](#) — Leave escalation reminder scheduler (every 8 hours)

#### Backup

- [0570ad0](#) — Database backup page with full export (including org\_positions and companies)
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## Day 2 — Apr 2: Seed Data (1 commit)

- [19254a1](#) — Seed org setup data and fix leave balance provisioning
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## Day 3 — Apr 3: RBAC & Kiosk (3 commits)

- [bc77a6e](#) — Fixed profile setup modal reappearing after photo captured
  - [d5091f9](#) — Multi-role RBAC system, employee self-service sections, kiosk improvements
  - [21ca8ff](#) — Home-page backup restore with bootstrap support, misc dashboard updates
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## Day 4 — Apr 4: Leave Accrual Engine & RBAC Migration (15 commits)

### BCEA Leave Accrual Engine

- [68a77d4](#) — Implemented BCEA-compliant leave accrual engine, fixed profile page
- [0dd9aa6](#) — Fixed double-counting of annual leave on backdated employee creation

### Leave UI

- [378fa84](#) — Leave request form validation improvements, show available balance
- [d94b576](#) — Fixed zero rendering artifact on leave balance cards
- [6ddc0b0](#) — Fixed stray '0' rendered when carryOverDays is 0
- [7f0d698](#) — Bypass React Query stale cache for fresh leave balances
- [5304dc3](#) — `formatLeaveDays` helper applied across displays; fixed manager filter
- [4ffe7fb](#) — Card-based leave type selector (replaced dropdown)
- [c3609b7](#) — Standard/Other leave balance grouping on employee dashboard

### RBAC Migration (legacy → `roles[]`)

- [92615c9](#) — Replaced `adminRole/userRole` with `roles[]` throughout auth and leave APIs
- [20a5f21](#) — Migrated OrgChart and MyProfileSection to `roles[]`
- [4f02895](#) — Migrated all role checks to `roles[]`, stripped passwords from API responses
- [80ab9d5](#) — Simplified reporting chain, hardened auth sessions, improved leave UI

### Dashboard

- [f10bb41](#) — Embedded OrgChart and AttendanceReports into dashboard sidebar layout
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## Day 5 — Apr 5: Deployment Hardening (12 commits)

### Deployment Tooling

- [2826673](#) — `.gitattributes` to enforce LF line endings for shell scripts
- [fa7950c](#) — Streamlined deployment tooling (Makefile, setup scripts)
- [9258995](#) — Strip CRLF from entrypoint during Docker build
- [5aa781f](#) — Run migrations via raw SQL/psql instead of drizzle-kit

## Backup/Restore Robustness

- [ffc9201](#) — Moved backup validation client-side (fix restore on new systems)
  - [0207287](#) — Fixed bootstrap detection, excluded face descriptors from backups
  - [a5b8565](#) — Stripped attendance photos from backup exports
  - [6535d8b](#) — Allowed bootstrap restore routes through auth middleware
  - [2b5780f](#) — Raw SQL count for bootstrap endpoints instead of `getAllUsers`
  - [ca3edfa](#) — Logged user import failures instead of swallowing silently
  - [40c50cb](#) — Revived date strings to Date objects during backup import
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## Day 6 — Apr 6: Leave UX Polish (1 commit)

- [f3ec00c](#) — Improved leave display, manager visibility, and rules dialog UX
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## Day 7 — Apr 7: Leave Projection & Half-Day Leave (12 commits)

### Major Features

- [d535500](#) — Future leave accrual projection engine
- [52e08ef](#) — Half-day leave support

### Leave Balance Fixes

- [8e3e3e6](#) — Fixed phantom carry-over, overlap detection, added in-app feedback
- [d03c1fc](#) — Balance check uses projected balance, cleaned up floating point display
- [1b82523](#) — Fixed balance indicator racing against projection load
- [c0571c8](#) — Deferred over-limit indicator to projection panel
- [16a7cf9](#) — Spinner while projection loads; use `projectedAvailable` once resolved
- [b424ca7](#) — Balance badge shows immediately, refines with projection when loaded
- [2c984cb](#) — Projected balance debug label in projection panel header
- [d65e935](#) — Fixed TDZ bug in projection, corrected available formula, improved trigger logic

### Documentation

- [e640ed9](#) — Updated SAD docs to reflect leave projection fixes
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## Day 8 — Apr 8: Theme System, Permissions & Feedback (7 commits)

### Feedback System

- [9b48c74](#) — Moved feedback button to header top-right next to theme toggle
- [1cdd69d](#) — Feedback icon button positioned furthest right, no label
- [3021778](#) — Button styled to match theme toggle (outline variant, icon size)
- [cdf1796](#) — Feedback/gateway env vars added to compose files

### Theme System

- `07d74aa` — Migrated all hardcoded palette colors to semantic theme tokens
- `bb6f74a` — Updated SAD docs to document semantic theme token system

Permissions & Config

- `bab0df2` — Configurable role permissions system
- `a55dd88` — Compose project names added, `.env.uat` ignored

Summary by Category

| Area                | Commi | Highlights                                                                      |
|---------------------|-------|---------------------------------------------------------------------------------|
| Leave System        | 25    | BCEA accrual engine, projection, half-day, custom rules, server-side validation |
| RBAC & Security     | 8     | <code>`roles[]`</code> migration, audit log, session auth, password stripping   |
| Docker & Deployment | 12    | Dockerfile, 4 compose files, Makefile, migration via psql, CRLF fixes           |
| Backup & Restore    | 8     | Bootstrap support, client-side validation, photo/descriptor exclusion           |
| UI/UX               | 7     | Semantic theme tokens, feedback button, card-based selectors                    |
| Permissions         | 1     | Configurable role permissions system                                            |
| Documentation       | 3     | README, SAD updates for projection and theme system                             |
| Misc Fixes          | 3     | Profile modal, seed data, org chart embedding                                   |

Feature Details

1. BCEA-Compliant Leave Accrual Engine

Files: `server/bcea.ts`, `server/leave-accrual.ts`

A pure-logic layer implementing South African Basic Conditions of Employment Act leave calculations, with all database access delegated to the accrual orchestration layer.

Core Capabilities:

- **Annual Leave Accrual** — Monthly accrual calculated as `rate x (activeDays / totalDays)`. Rate resolved via priority chain: per-employee override > highest qualifying tenure tier > fallback (15 days/yr). No rounding; exact decimals stored per spec.
- **Graduated Sick Leave (First 6 Months)** — Accrues 1 day per 26 days worked during the probationary period, with cumulative counters for idempotency. At the 6-month mark, transitions to full entitlement: `30 x (workDaysPerWeek / 5)` minus days already taken.
- **Family Responsibility Leave** — 3-day grant at the 4-month mark, conditional on `completedMonths >= 4` and `workDaysPerWeek >= 4`.
- **Accrual-Pausing Leave Types** — Unpaid, Maternity, Parental, Adoption, and Commissioning leave days are deducted from active days in a month, reducing pro-rated accrual.
- **Carry-Over** — Configurable grace period (default 6 months after annual cycle end). Expired carry-over is forfeited.

- **Backfill** — When an employee is created with a past start date, all completed months are backfilled idempotently via unique constraint on (`employeeId`, `leaveType`, `accrualPeriod`, `eventType`).
- **Termination Settlement** — Pro-rated accrual for mid-month termination with immediate balance update and HR notification.

**Statutory Entitlements:** Maternity (87 days), Parental (10), Adoption (50), Commissioning (50).

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## 2. Custom Leave Rules Engine

**File:** `server/custom-leave-rules.ts`

Handles non-BCEA leave types (e.g., study leave, religious leave) via a configurable rule/phase system that runs alongside the BCEA engine.

- **Phase-Based Entitlement** — Each rule can have multiple phases with different accrual settings that activate at tenure milestones. The engine resolves the most advanced phase the employee qualifies for.
  - **Four Accrual Types:**
    - `per_days_worked` — `floor(approx_working_days / periodDaysWorked) x daysEarned`
    - `monthly` — `daysEarned x totalMonths`
    - `annual` — `daysEarned x completedYears`
    - `fixed_per_cycle` — flat allocation per cycle
  - **Max Accrual Cap** — Optional ceiling per rule. Values rounded to 1 decimal place.
  - Only updates existing balances — HR/Admin must manually activate leave types for employees.
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## 3. Future Leave Accrual Projection

**File:** `server/leave-projection.ts`

Advisory-only engine that simulates future balances without writing to the database. Used to show employees what their balance will be at a future date.

**Algorithm:**

1. Loads current balances, leave rules, holidays, accrual tiers, and active requests
2. Builds month range from last accrued month through to the target date
3. For each month: determines tenure-based accrual rate, counts pausing leave days, calculates pro-rated accrual via BCEA formula
4. Detects annual cycle resets and computes carry-over
5. Simulates settlement of pending requests with start dates before the target date

**Output:** `currentAvailable`, `projectedAvailable`, `projectedAccrual`, `cycleResetOccurs`, `carryOverCreated`, `carryOverExpiry`, `monthsProjected`, and a per-month `breakdown[]`.

**UI Behaviour:** Balance badge shows immediately with the current value, then refines with the projected value once the calculation completes. A spinner indicates loading. The over-limit indicator is deferred to the projection panel when active.

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## 4. Half-Day Leave

**Schema:** `startHalfDay` and `endHalfDay` columns (enum: `'AM' | 'PM' | null`).

- Each half-day flag reduces the requested total by 0.5 days (e.g., 3 working days with an AM start = 2.5 days)

- Single-day requests use `startHalfDay` only; `endHalfDay` is ignored when `startDate === endDate`
- Half-day adjustments carry through pending-to-taken settlement transitions
- UI: Toggle buttons (AM/PM) appear after date selection; single-day mode hides the end half-day option

## 5. Multi-Role RBAC System

**Schema:** `roles: text[]` on the users table, replacing legacy `role` and `adminRole` fields. Explicit set: `employee`, `manager`, `hr`, `admin`. Roles are additive with no implicit inheritance.

**Migration:** All auth checks, leave APIs, OrgChart, MyProfileSection, and API responses were migrated from the legacy single-role fields to the `roles[]` array. Passwords are stripped from all API responses.

## 6. Configurable Role Permissions

**File:** `client/src/hooks/use-role-permissions.ts`

Runtime-configurable permission system stored in the `settings` table under key `role_permissions`.

**Hook API:**

```
const { canSee, canDo } = useRolePermissions();
canSee('nav.personnel') // visibility check
canDo('personnel.terminate') // action check
```

- **Admin bypass** — Admin role always returns true for all checks.
- **Non-admin** — Returns true if any of the user's roles has the permission in the map.
- **Default permissions** define what each role can see and do out of the box (e.g., employees see leave/attendance/profile; managers see team/org-chart/reports; HR sees personnel management actions).
- **Admin UI** — Checkbox grid in Settings (roles x permissions) for runtime configuration.
- **Applied to:** Dashboard nav items (`canSee`), PersonnelSection actions (`canDo` for add, export, edit, terminate, delete, assign roles).

## 7. Docker & Deployment Architecture

**Files:** `Dockerfile`, `docker-compose.yml`, `docker-compose.dev.yml`, `docker-compose.uat.yml`, `docker-compose.release.yml`, `Makefile`, `setup.sh`, `docker-entrypoint.sh`, `docker-entrypoint.dev.sh`

**Multi-Stage Dockerfile:**

1. **Builder** — Compiles TypeScript to `dist/`, handles native modules (bcrypt, pg)
2. **Runner** — Alpine-based, copies compiled artifacts, runs entrypoint

**Three Deployment Profiles:**

| Profile                | Compose File                              | Details                                                                          |
|------------------------|-------------------------------------------|----------------------------------------------------------------------------------|
| <b>**Development**</b> | <code>`docker-compose.dev.yml`</code>     | Source mounted for live reload, <code>`npm run dev`</code> with HMR              |
| <b>**UAT**</b>         | <code>`docker-compose.uat.yml`</code>     | Separate ports (5002), passwords, and data dirs to avoid prod collision          |
| <b>**Production**</b>  | <code>`docker-compose.release.yml`</code> | Pre-built image from <code>`ghcr.io/stopdoingnothing/factory-flow:latest`</code> |

**Common Services:**

- PostgreSQL 16-Alpine with health checks



- Automated backup service: `pg_dump` every 6 hours, 90-day retention

#### Entrypoint (production):

1. Waits for Postgres TCP ready
2. Applies migration SQL files (stripping Drizzle markers)
3. Pre-creates `sessions` and `audit_logs` tables
4. Executes `node dist/index.cjs`

**Makefile:** `setup`, `start`, `stop`, `restart`, `status`, `logs`, `update` (git pull + rebuild), `backup`, `restore` (interactive), `reset-data` (with safety prompt).

**setup.sh:** First-run wizard that prompts for Postgres password, app URL, optional Postmark key, generates `SESSION_SECRET`, and writes `.env`.

## 8. Backup, Restore & Bootstrap

**Location:** `server/routes.ts` (bootstrap endpoints)

**Bootstrap Detection:** `GET /api/backup/bootstrap-check` returns `{ empty: true }` on a fresh database (catches undefined table error). Allowed through auth middleware so a new deployment can be restored without credentials.

#### Bootstrap Import Flow:

1. Validates JSON structure and counts entities
2. Rejects if any users already exist
3. Revives ISO date strings to Date objects (lost during JSON serialization)
4. Imports in dependency order: departments → userGroups → employeeTypes → companies → orgPositions → users → leaveBalances → remaining entities
5. Idempotent — checks for existing records before creating

**Excluded from Exports:** Face descriptors (large binary data), attendance photos (storage-heavy), audit logs (system-generated), session records (ephemeral).

**Automated Backups:** `pg_dump` via backup service every 6 hours to `./data/backups/factoryflow_YYYYMMDD_HHMMSS.sql.gz`. Files older than 90 days are pruned.

## 9. Audit Log

**Schema** (`shared/schema.ts`):

| Column                     | Type  | Purpose                                                                                                    |
|----------------------------|-------|------------------------------------------------------------------------------------------------------------|
| <code>`actor_id`</code>    | text  | Session userId; null for system actions                                                                    |
| <code>`action`</code>      | text  | e.g., <code>`update_user_profile`</code> , <code>`update_leave_balance`</code>                             |
| <code>`entity_type`</code> | text  | <code>`user`</code> , <code>`leave_balance`</code> , <code>`setting`</code> , <code>`leave_request`</code> |
| <code>`entity_id`</code>   | text  | Primary key of affected row                                                                                |
| <code>`changes`</code>     | jsonb | <code>`{ field: { before, after } }`</code> or event payload                                               |
| <code>`description`</code> | text  | Human-readable one-liner                                                                                   |

Pre-created at startup by both entrypoint scripts. Covers: user profile updates (sensitive field diffs), leave balance manual adjustments, leave request approvals/rejections, and settings changes. Paginated fetch via `GET /api/audit-logs` (admin only).

## 10. Semantic Theme Token System

**File:** `client/src/index.css`

CSS custom properties in HSL format defined at `:root` (light) and `.dark` class, mapped to Tailwind via `@theme`.

**Palette:** Dark red primary (`#8B1A1A`), near-black foreground, light grey backgrounds. Status tokens for success (green), warning (orange), info (blue), destructive (red) — each with a `-muted` variant for backgrounds.

**Utility Classes:** `.glass-panel` (frosted glass effect), `.industrial-card` (left-border accent), `.btn-industrial` (uppercase, tracking, scale on active).

All previously hardcoded palette colors across the codebase were migrated to these semantic tokens, ensuring consistent theming and dark/light mode support throughout.

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## 11. In-App Feedback System

**Files:** `client/src/components/FeedbackModal.tsx`, `client/src/lib/feedback.ts`, `server/routes.ts`

**UI:** Modal with tabs for "Bug Report" (title, severity, steps, actual/expected behaviour) and "Feature Request" (title, motivation, proposed behaviour). Captures the current route automatically. Styled as an icon button in the header next to the theme toggle.

**Client-Side Resilience:**

- Pending reports stored in `localStorage` (key: `ff_pending_feedback`)
- Idempotency key: hash of `type:title:route` + timestamp
- Exponential backoff retry (up to 3 attempts, max 48-hour persistence)
- `flushPendingFeedback()` retries unsent reports on app start

**Server Integration:** `POST /api/feedback` forwards to the Agent Gateway, which creates GitHub Issues in the configured repository. Returns `{ success, issue_url, issue_number }`. Gracefully degrades (503 if not configured, 502 if gateway fails).

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## 12. Server-Side Leave Validation & Hardening

Additions that moved validation logic from the client to the server:

- **Statutory leave types** enforced server-side
- **Unpaid leave** 7-day notice validation with admin discretion bypass
- **Medical certificate** flag auto-triggered on sick leave submission; upload enforced before HR can approve
- **Leave overlap detection** prevents double-booking
- **Escalation reminders** run every 8 hours for pending approvals
- **HR approval stage** made optional via `leave_require_hr_stage` setting
- **HR reporting endpoint** (`GET /api/reports/leave`) for aggregated leave data
- **Carry-over grace period** configurable via settings (default 6 months)